

Best Practices for Intrusion Detection and Threat Mitigation in Cloud Environments

by

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Declaration

I, *Banele Mlonyeni, 224612255i*, hereby declare that the *Research Report for the Postgraduate Diploma in Cybersecurity* is my own work and that it has not previously been submitted for assessment or completion of any postgraduate qualification to another University or for another qualification.

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Abstract

Cloud computing has transformed how organisations store, process, and secure data by offering flexibility, scalability, and cost efficiency. However, this rapid adoption has introduced new security challenges, especially in detecting and responding to threats within complex and decentralised environments. This study addresses the problem of the slow adaptability of intrusion detection and threat mitigation strategies to the evolving nature of cloud environments, resulting to delayed detection, increased false positives, and high risks of cyberattacks and unauthorised access. The purpose of this study is to develop best practices for enhancing intrusion detection and threat mitigation in cloud environments.

A qualitative approach was adopted, involving a narrative literature review (SO1) to identify existing challenges and limitations, and a systematic literature review (SO2) to identify effective mechanisms. A semi-structured interview (SO3) was designed to guide future primary data collection from cybersecurity professionals to validate the proposed practices. Objectives SO1 and SO2 were achieved through comprehensive literature synthesis, while SO3 will form part of future research.

Findings indicate that current IDS implementations face significant challenges, such as scalability constraints, false positive rates, adaptability, and limited visibility across distributed systems. Machine learning-driven IDS approaches enhance detection accuracy and adaptability in cloud environments. Automated response mechanisms, including orchestration and policy-based control, support improved real-time mitigation. The study contributes by consolidating best practices and providing a foundation for a practical framework to strengthen intrusion detection and response in cloud environments.

Keywords

Adaptive Security, Best Practices, Cloud Computing, Cloud Security, Intrusion Detection Systems (IDS), Machine Learning, Threat Detection

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Acronyms

ACRONYM	DESCRIPTION
ACIRS	Autonomous Cloud Intrusion Response System
AI	Artificial Intelligence
APT	Advanced Persistent Threat
BPSO	Binary Particle Swarm Optimisation
CNN	Convolutional Neural Network
CSP	Cloud Service Provider
DDoS	Distributed Denial of Service
DL	Deep Learning
DNN	Deep Neural Network
ELSTM	Enhanced Long Short-Term Memory
GDPR	General Data Protection Regulation
GNN	Graph Neural Network
HIPAA	Health Insurance Portability and Accountability Act
IAM	Identity and Access Management
IDS	Intrusion Detection System
IDSaaS	Intrusion Detection System as a Service
IDSGT-DNN	Intelligent Deep Security Game Theory – Deep Neural Network
IPS	Intrusion Prevention System
ISO	International Organisation for Standardisation
IT	Information Technology
KNN	K-Nearest Neighbour
ML	Machine Learning

MLPNN	Multi-Layer Perceptron Neural Network
NIDS	Network Intrusion Detection System
NIST	National Institute of Standards and Technology
NSL-KDD	Network Security Laboratory – Knowledge Discovery in Databases
OSVM	One-Class Support Vector Machine
POPIA	Protection of Personal Information Act
PRO	Primary Research Objective
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PSO	Particle Swarm Optimisation
REC-H	Research Ethics Committee – Humanities
RF	Random Forest
RNN	Recurrent Neural Network
ROI	Return on Investment
SCAE	Stacked Contractive Autoencoder
SDN	Software-Defined Networking
SLR	Systematic Literature Review
SO	Secondary Objective
SSIR	Semi-Supervised Intrusion Recognition
SVM	Support Vector Machine
VM	Virtual Machine
WSNs	Wireless Sensor Networks
ZTA	Zero Trust Architecture

Chapter 1 - Introduction

1.1. Theoretical Background / Related Literature

1.1.1. Cloud Security

Cloud computing has evolved as a transformative technology, enabling scalability and flexibility, and changing how data and services are delivered across industries (Pali et al., 2023). Organisations store sensitive data and run mission-critical applications on these platforms; therefore, secure cloud environments are critical (Sharma & Sharma, 2024).

Cloud security refers to the collection of technologies, policies and procedures that ensures safety of cloud-based systems, data and infrastructure. To ensure confidentiality, integrity, and availability is its primary objective (Ahmadi, 2024a).

Unlike traditional IT systems which are managed at fixed location, cloud environments are dynamic and decentralised (Prasad, 2024). This means that resources can be managed by third-party, shared across different users, and moved between location (Ahmadi, 2024a). However, securing these systems has become a new challenge with this growth (Pali et al., 2023). Key concerns include limited visibility, shared responsibility, multi-tendency, and real-time provisioning (Bhumika et al., 2024) (Bhumika et al., 2024). Among other things, these features increase cloud systems resilience but making them difficult to secure. For example, ensuring visibility across all systems becomes difficult, also difficult to monitor user activities, and responding to threats in real-time (Pali et al., 2023).

While using the cloud, organisations face many types of threats such as data breaches, unauthorised access, insider threats, and denial of service attacks (Ahmadi, 2024b). Reports show that several cyberattacks targeting cloud infrastructures has increased by more than 150 percent in the recent years, making flexible security posture crucial (Pali et al., 2023).

1.1.2. Intrusion Detection

Intrusion Detection Systems (IDS) are a crucial defence mechanism that monitors network activity for signs of malicious behaviour (Jimmy, 2024; Zaidi, 2021). Signature-based IDS works by matching incoming traffic patterns against a database of known attack (Zaidi, 2021). They are effective in detecting known threats, but fail to detect zero-day or novel attacks that do not match any existing signatures, leaving systems vulnerable to new threats (Prasad, 2024; Pali et

al., 2023). On the other hand, anomaly-based IDS operate by detecting unusual behaviour that does not align with the known patterns of activity and therefore can detect novel attacks (Zaidi, 2021). However, anomaly-based IDS suffer from false positives, where legitimate traffic is detected as malicious traffic, resulting in reduced trust in the system and alert fatigue (Lata & Singh, 2022).

To enhance the accuracy and reduce the number of false alarm issues, hybrid IDS models integrate the benefits of signature-based and anomaly-based systems (Ahmadi, 2024a; Jimmy, 2024). For proactive detection, this development has machine learning that can now be used in IDS systems.

1.1.3. Theoretical and Framework Foundations

The theoretical frameworks such as Intelligent Deep Security Game Theory – Deep Neural Network (IDSGT–DNN) model use both cooperative and non-cooperative game structures to simulate the interactions between attackers and defenders (Paramesh et al., 2024). Cloud infrastructures' strategic defence mechanisms is improved by this approach (Paramesh et al., 2024).

Additionally, this approach enable the integration of REST APIs and microservices, enhancing the effectiveness of scalability and compatibility with existing cloud systems (Jimmy, 2024; Jyothsna et al., 2025). These models in dynamic corporate environments promote the adaptability to evolving business threats to ensure smooth business continuity (Murali et al., 2024).

Researchers highlight the significance of public datasets, such as NSL-KDD and CICIDS2017 (Balisane et al., 2024). Detection accuracy, response time, and false positives/negatives are common metrics for evaluation. Additionally, there is growing trend to use multi-cloud approaches to reduce the risk of relying on single cloud provider and enhance overall system resilience (Balamurugan et al., 2022).

1.1.4. Machine, Hybrid Techniques, and SDN

Machine learning integrate probabilistic models which predict uncertainty and likelihood, showing a significant improvements in detection accuracy (Prasad, 2024; Paramesh et al., 2024). Detection accuracy and reduced false-positives alerts is improved by these frameworks that combine signature-based and anomaly-based approaches (Pali et al., 2023).

The use of deep learning algorithms, including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks improves detection of complex patterns in network traffic. They

enable real-time detection and dynamic responses to zero-day attacks (Murali et al., 2024).

Other tool that is being explored to reduce Distributed Denial of Service (DDoS) attacks is Software Defined Networking (SDN). SDN adds a programmable layer of defence to enable the ability to dynamically re-route network traffic. SDN can provide real-time threat detection and proactive protection when integrated with AI and ML (Murali et al., 2024; Paramesh et al., 2024).

1.1.5. Socio-Technical and Policy Considerations

Successful deployment of IDS is likely to be prevented by socio-technical factors such as user acceptance and organisational culture despite adoption of technology (Ahmadi, 2024a). Ability to change security configurations and user control is limited by cloud transparency issues. Additionally, the need for strong audit trails and accountability frameworks is emphasised by the legal and compliance issues (Ahmadi, 2024a).

NISP SP 800-53 and ISO 27001 are global standards that are important in guiding organisations towards best practices. However, these standards must evolve along with technology improvements and integrate adaptive techniques to continue being effective in dynamic cloud environments (Ahmadi, 2024a; Bhumika et al., 2024).

1.1.6. Research Gaps and Future Directions

Real-world case studies that examine how organisations address cloud security incidents is lacked by current literature (Balamurugan et al., 2022). Research on emerging technologies, such as quantum computing and how they impact IDS is also limited (Jyothsna et al., 2025).

The use of transfer learning, integrated approaches, and advanced feature engineering should be examined by future research to develop more robust models (Bhumika et al., 2024). Additionally, cloud environments high traffic volume requires scalable and effective IDS architectures (Pali et al., 2023). The development of practical and implementable IDS solutions continues to prioritise the balance between accuracy of detection with computing cost and usability (Ahmadi, 2024b).

1.2. Description of Problem

Cloud computing is now a significant approach for modern technology (Pali et al., 2023). Organisations use cloud services to store data, run applications, and rapidly scale their systems based on the current needs. Benefits such as efficiency improvement, enhanced flexibility, and cost-effectiveness are acquired using this approach (Bhumika et al., 2024). As a result, its

adoption is used by many industries, including education, finance, government, and healthcare. However, this growing reliance on the cloud has created a significant new security challenges (Pali et al., 2023).

Organisations face many threats, including insider threats, data breaches, unauthorised access, and denial of service attacks while using cloud (Ahmadi, 2024b). Intrusion detection systems are crucial to detect these threats, which are designed to monitor systems and alert administrators about suspicious activities (Ahmadi, 2024b). However, many traditional IDS approaches are designed for older environments with static systems, not for today's rapidly evolving cloud platforms (Prasad, 2024).

Signature-based IDS match patterns against a database of known threats to detect attacks (Lata & Singh, 2022). In cases where an attack has been detected already before, signature-based are effective. However, these systems are ineffective when it comes to detecting novel attacks, such as zero-day attacks (Lata & Singh, 2022).

On the other hand, anomaly-based IDS are effective when detecting unknown patterns of behaviour (Lata & Singh, 2022). Although these systems are effective when detecting unknown threats, they suffer from false positives, resulting in confusion, time wasting, and reduced detection system trust (Balisane et al., 2024). Neither approach operates well on its own when implemented in complex environments, both have significant vulnerabilities.

The problem becomes more complicated when addressing the unique features of cloud computing (Paramesh et al., 2024). Usually, in the cloud, different providers that manage virtual machines, networks, and services (Prasad, 2024). There is limited visibility into these domains. Attackers can move through these systems without being detected by taking advantage of these gaps, escalating their access and compromise confidential information. It is difficult to detect insider threats, for example when someone inside the organisation misuses access for harm because traditional systems are not designed to handle that level of complexity (Pali et al., 2023).

Researchers and practitioners have started exploring new techniques to make cloud systems more secure. These techniques, include using hybrid IDS systems that combine signature-based and anomaly-based methods to strengthen systems (Prasad, 2024). These systems aim to improve accuracy and reduce false positives. Additionally, machine learning and deep learning are also being used to help systems "learn" from the past data and identify patterns that might indicate an attack (Bhumika et al., 2024; Murali et al., 2024). Automated response systems make it possible to respond immediately when a threat is detected, reducing damage well

before human action becomes possible (Paramesh et al., 2024).

Many organisations still have no idea on how to use them appropriately in real-world although these sophisticated solutions sound promising. There isn't a cohesive set of best practices that put together these ideas into a usable, tested guide. Security experts are pretty much left to figure out on their own, usually without enough evidence or support. Additionally, there is lack of collaboration between cloud providers, customers, and cybersecurity experts to develop security rules and recommendations (Ahmadi, 2024b; Pali et al., 2023).

As a result, many organisations have outdated, inconsistent, or poorly implemented intrusion detection systems (Pali et al., 2023). Lack of consistent industry standards along with the absence of accurate guidance increase the risk of successful cyberattacks, data breaches, and financial losses (Ahmadi, 2024b). If there is no shared understanding of what works best in different cloud environments, the security of critical systems and information remains at risk (Pali et al., 2023).

This calls for a collection and evaluation of the most effective strategies to detect and respond to cloud-based cyber threats. This involves examining current intrusion detection approaches, identifying gaps in existing methods, and learning from expert opinions. The aim is to develop clear, evidence-based best practices that organisations can use to improve their security posture in the cloud and to reduce risk.

1.3. Problem Statement

Intrusion detection and threat mitigation strategies are slow to adapt in evolving nature of cloud environments, resulting in delayed detection, increased false positives, and high risk of cyberattacks and unauthorised access.

1.4. Research Objectives

Primary Objective:

To develop best practices for enhancing intrusion detection and threat mitigation strategies in cloud environments.

Secondary Objectives:

SO1 (Literature Review):

To identify current challenges and limitations of intrusion detection systems (IDS) in cloud environments.

SO2 (Systematic Literature Review):

To identify effective intrusion detection systems and automated threat response mechanisms implemented in cloud environments.

SO3 (Interview):

To obtain cybersecurity professional's perspectives and recommendations on intrusion detection systems and response mechanisms in cloud environments.

1.5. Research Methodology

This study is using qualitative method to meet research objectives. Critical reasoning is applied to synthesise findings across different methods of the study. These methods include narrative literature review to identify challenges, a systematic literature review to identify effective practices, and semi-structured interviews to obtain expert perspectives and recommendations as shown in the Table 1.1. Implementing this multi-method approach supports methodological triangulation, enhancing the validity and reliability of the research findings. The development of well-informed best practices for intrusion detection and threat mitigation in cloud environments is allowed by the integration of insights from literature, analysis, and professional input.

Table 1.1 Research Objectives, Related Methods, and Envisaged Output

Research Objective	Research Method	Envisaged Output/Deliverable
PRO: To develop best practices for enhancing intrusion detection and threat mitigation strategies in cloud environments.	Critical Reasoning	A set of documented and validated best practices for intrusion detection and threat mitigation in cloud environments.
SO1: To identify current challenges and limitations of intrusion detection systems (IDS) in cloud environments.	Literature Review	A synthesised list of key challenges and limitations of IDS in cloud environments.
SO2: To identify effective intrusion detection systems and automated threat response mechanisms implemented in cloud environments.	Systematic Literature Review	A summary of effective intrusion detection systems and automated threat response mechanisms implemented in cloud environments.
SO3: To obtain cybersecurity professional's perspectives and recommendations on intrusion detection systems and response mechanisms in cloud environments.	Semi-structured Interview	A thematic summary of expert perspectives and recommendations on intrusion detection systems and response mechanisms in cloud environments.

Research Methods Descriptions

Critical Reasoning: Refer in Section 4.4 of Chapter 4 for detailed explanation.

Literature Review: According to Liang and Xu (2025) and Snyder (2019), literature review synthesises existing knowledge to establish what is already known and where gaps exist. Refer to Section 4.2 of Chapter for more detailed discussion.

Systematic Literature Review: SLR is a method for identifying literature through a predefined protocol. It emphasises methodological, transparent, and rigor (De Souza et al., 2022a; Van Dinter et al., 2021a). Refer to Section 4.2 of Chapter for more detailed discussion.

Semi-structured Interview: According to Adeoye-Olatunde and Olenik (2021), semi-structured interviews are valuable when exploring under-researched or complex topics because they allow participants to elaborate beyond initial questions. Refer to Section 4.2 of Chapter for more detailed discussion.

Sampling: A combination of convenience and purposive sampling will be used by this study. Refer in Section 4.4 of Chapter 4 for more detail.

1.6. Ethical considerations

This study will adhere with Nelson Mandela University Research Ethics Committee-Human (REC-H) guidelines to ensure participant protection, confidentiality and anonymity, and data security. A detailed explanation of the ethical procedures and safeguards implemented in this study is discussed in more detail in Section 4.6 of Chapter 4.

1.7. Chapter Layout

This research report is structured into five chapters, each addressing a specific component of the study. The chapters are outlined and briefly described their purpose and content in the Table 1.2.

Table 1.2: Chapter Layout

Chapter	Chapter Title	Description and Purpose of Chapter
Chapter 1	Introduction	Introduces the research topic, background and context, problem statement, research objectives, and significance of the study. It also outlines the research methodology overview, ethical considerations, and concludes with a layout of the remaining chapters.
Chapter 2	Literature Review	Critically reviews and analyses existing literature to address SO1, which is to identify current challenges and limitations of IDS in cloud environments. It explores academic and industry studies emphasising technical, organisational, operational, and legal factors that affect IDS performance. This chapter also identifies research gaps and recurring issues in existing approaches.
Chapter 3	Systematic Literature Review	Describes the systematic search strategy, inclusion and exclusion criteria, and thematic analysis process. This chapter consolidates findings from credible sources such as ScienceDirect, IEEE Xplore, and Scopus to identify effective strategies and emerging trends to address SO2.
Chapter 4	Research Design	Outlines the research methodology adopted to address SO3. It details the research process, data collection strategy, participant sampling, and instrument design. This chapter also discusses ethical considerations which adhere Nelson Mandela University REC-H guidelines.
Chapter 5	Conclusion	Summarises the entire research by revising the objectives and evaluating how each was achieved. Emphasises the key findings, practical implications, limitations, and recommendations for future work. Reflects on the research process and provides final remarks on the study's contribution to cybersecurity and cloud security practices.

Chapter 2 - Literature Review

2.1. Introduction

The purpose of this chapter is to review and critically analyse existing literature to address the SO1 of this study, which is “to identify current challenges and limitations of intrusion detection systems (IDS) in cloud environments”. As discussed in Chapter 1, cloud computing is dynamic, decentralised, and multi-tenant unlike traditional IT systems which are managed at fixed location. However, this raise security concerns which traditional intrusion detection approaches were not designed to handle. These challenges compromise the effectiveness of existing security approaches, leaving cloud environments vulnerable to cyberattacks.

To achieve this objective, the literature review examines academic research, industry reports, and recent studies focusing specifically on the operational, technical, and organisational factors that contribute to the challenges and limitations of implementing IDS in the cloud. The analysis aims to identify recurring issues, analyse the causes, and recognising gaps in current approaches that compromise the effectiveness of intrusion detection in cloud settings.

Relevant literature was identified through a structured search in databases such as ScienceDirect, IEEE Xplore, and Scopus. The search made use of keyword combinations such as “anomaly-based IDS,” “cloud security challenges,” “hybrid intrusion detection cloud,” “IDS limitations cloud,” “intrusion detection systems in cloud computing,” and “signature-based IDS.” To ensure relevance and quality, only peer-reviewed articles, academic publications, and reliable technical reports published between 2020 and 2025 were included.

This chapter is organised as follows. Section 2.2 introduces IDS in cloud environments. Section 2.3 examines the key limitations of IDS in cloud environments, while Section 2.4 discusses the challenges of implementing IDS in cloud environments. Finally, Section 2.5 concludes with a summary of findings and their significance to this study’s objectives.

2.2. Overview of Intrusion Detection Systems (IDS) in Cloud Environments

2.2.1. Definition and Role of IDS in Cloud Security

Intrusion Detection Systems (IDS) are a type of specialised security tool that actively monitors the network and system activities for signs of malicious behaviour or policy violations (Alnasser et al., 2025). They play a crucial role by maintaining system integrity to detect unauthorised access attempts, protects sensitive data, detect malicious activities, or system anomalies in cloud

environments (Ahmed, 2024; Almuseelem, 2025; Alnasser et al., 2025). They examine network traffic by capturing and decoding the network traffic, reassembling TCP streams, and checking for patterns or anomalies indicative of potential threats (Prasad, 2024). IDS can prevent tampering of logs by attackers, and they also detect known attack patterns by working in a passive mode. However, Riaz et al., (2017) and Prasad (2024) stated that traditional IDS models often struggle when it comes to dynamic cloud environments where attack vendors continually evolve and the lack of fixed infrastructure complicate detection. While literature consistently emphasises IDS as a fundamental security layer, much of the focus remains on general monitoring functions. There is a gap in research on how IDS log management and passive monitoring adapt to virtualised, multi-tenant cloud infrastructures.

2.2.2. IDS Detection Tools

IDS technologies are classified into three categories, as signature-based, anomaly-based, and hybrid systems (Alhayan et al., 2025). Signature-based IDS compares incoming traffic with databases of known attack signatures, making them effective at detecting known threats, but ineffective at detecting zero-day attacks (Ahmadi, 2024a; Lata & Singh, 2022). On the other hand, Shadi Alhakeem and Bin Ajla (2024) stated that anomaly-based IDS operate by detecting unusual behaviour that does not align with the known patterns of activity and therefore can detect novel attacks. They suffer from high false positives although they detect zero-day attacks, which can lead to alert fatigue and reduced trust in systems (Riaz et al., 2017; Dina & Manivannan, 2021). Hybrid IDS approaches combine the benefits of both methods to provide for efficient communication between known-pattern identification and anomaly detection, which helps to improve accuracy and reduce false alarms (Alnasser et al., 2025; Shadi Alhakeem & Bin Ajla, 2024). Research shows that hybrid IDS is a solution to the individual problems of signature-based and anomaly-based models. However, many proposed hybrid frameworks lack large-scale deployment evidence in real-world cloud environments.

2.2.3. Traditional IDS versus Cloud-based IDS

There is significant difference between traditional IDS on-premises IT infrastructure and IDS in cloud computing. Cloud environments are more dynamic, decentralised, and multi-tenant, requiring the need for sharing resources across clients (Ahmed, 2024; Prasad, 2024). These characteristics pose challenges, including the need to handle large volumes of high dimensional data, monitor traffic across distributed architectures, and ensure effective detection without compromising performance (Lata & Singh, 2022). On the other hand, traditional IDS are more static and operate in more predictable network environments (Riaz et al., 2017). Additionally, integration of other multi-tenant

configurations, such as third-party services, virtual machines, and multi geographies should be handled by cloud IDS because they increase attack surface (Binbusayyis, 2024; Mayuranathan et al., 2022). Very few studies seem to develop tailored comprehensive frameworks to hybrid or multi-cloud scenarios although most studies acknowledge the operational differences between traditional and cloud IDS. This is a drawback for businesses with a variety of cloud service models because there are not many effective frameworks that are specifically customised.

2.2.4. Importance of IDS in Protecting Against Cloud Threats

Detecting various sophisticated threats such as account hijacking, insecure API usage, insider threats, ransomware, and distributed denial-of-service (DDoS) attacks is one of the most significant aspects of IDS in the context of cloud security (Alhayan et al., 2025; Mayuranathan et al., 2022). Timely alerts and automated or semi-automated cloud service response systems ensure the protection of cloud services. According to Almuseelem (2025) and Prasad (2024), integrating machine learning (ML) and artificial intelligence (AI) has significantly enhanced IDS frameworks in the cloud, making them more adaptable, scalable, and accurate. Despite benefits that AI/ML brings, current research lacks consensus on standardised training datasets and evaluation benchmarks, which makes comparing different IDS solutions challenging. Without these standards, it is nearly impossible to validate performance improvements in real-world scenarios.

2.3. Key Limitations of IDS in Cloud Environments

Intrusion Detection Systems (IDS) deployed in cloud environments encounter several limitations that reduce their operational effectiveness. These limitations can generally be grouped into four categories: (1) signature-based IDS, (2) anomaly-based IDS, (3) hybrid IDS, and (4) limitations that are unique to cloud settings. Recognising these challenges is significant, as it not only emphasises areas where current systems fall short but also provides direction for developing solutions that are more adaptive and scalable.

2.3.1. Signature-Based IDS Limitations

Signature-based IDS works by matching incoming traffic patterns against a database of known attack signatures or predefined rules (Alnasser et al., 2025). This approach operates effectively for previously detected threats, but it cannot detect zero-day or new attacks that do not yet have signatures, which exposes systems to emerging threats (Ahmadi, 2024a; Almuseelem, 2025). In practice, attackers can avoid detection by using simple obfuscation methods such as encoding or encryption, and even few modifications to malicious code may avoid detection (Ahmadi, 2024a). Frequently updated and comprehensive signature databases result in increased computational overhead and storage (Riaz

et al., 2017; Lata & Singh, 2022). Signature-based IDS are reactive rather than proactive regardless of having high accuracy for known threats because of their resilience on current signature databases. The absence of scalable methods for updating signatures in real time over distributed multi-tenant cloud infrastructures is a significant research gap.

2.3.2. Anomaly-Based IDS Limitations

Anomaly-based IDS monitor deviations from norms, providing the ability to detect unknown threats (Alnasser et al., 2025). The advantage of this approach is that it enables the detection of new and unknown threats (Alnasser et al., 2025; Prasad, 2024). However, Alnasser et al. (2025) found that this approach suffers from high rate of false positives, where normal traffic is identified as malicious. These false alarms significantly overwhelm security personnel, resulting in alert fatigue. Additionally, anomaly detection consumes high amounts of processing resources, which is an issue in large-scale, multi-cloud environments (Zhang et al., 2024). A data imbalance that significantly outweighs malicious samples can also result in false negatives, which reduces the accuracy of detection (Binbusayyis, 2024). The operational costs and persistent issues of dealing with false positives limit the anomaly-based IDS accuracy regard the ability to detect novel threats. To reduce false alarms, emerging trends suggest integrating anomaly detection with AI-driven feature selection, nevertheless, most of systems are still in the experimental stage and need reliable cloud-scale validations.

2.3.3. Hybrid IDS Limitations

Hybrid IDS model combines anomaly-based and signature-based approaches to integrate their benefits (Alhayan et al., 2025). Although theoretically sound better, Dina and Manivannan (2021) stated that this method increases integration complexity and adjusting parameters. Especially when used across virtualised environments, managing many detection models at once can result in higher computational costs, latency, and system overhead (Alhayan et al., 2025). Moreover, maintaining model relevancy requires continuous retraining, which is expensive in terms of resources and might reduce performance if not optimised (Zhang et al., 2024). Hybrid IDS is becoming increasingly common in both academic and industry research. Lack of standardised integration frameworks is one of the major challenges on implementing multi-cloud hybrid models.

2.3.4. Cloud-Specific Limitations

Cloud environments pose unique challenges that traditional IDS were not designed to handle because they are less adaptable (Ahmadi, 2024a). Key challenges include limited visibility over virtual machines (VMs) and hypervisors, the difficulty of analysing encrypted traffic, and the demand for

flexibility as workloads constantly change (Riaz et al., 2017; Mayuranathan et al., 2022). In hybrid setups, constraints in resource allocation can further compromise IDS performance, and this is combined with the lack of transparency into the cloud provider's own security operations, leaving organisations with potential security gaps (Hasan, 2024). Additionally, to perform real-time monitoring and analysis across different systems is harder because of distributed nature of cloud infrastructure. These cloud-specific limitations emphasise the need for IDS solutions that interact with cloud management systems and enable analysis of encrypted traffic. About this multi broad design issue, there are few solutions that attempt to solve it and there is still a little research in the field.

2.4. Challenges in Implementing IDS in Cloud Environments

The implementation of Intrusion Detection Systems (IDS) in cloud environments is impacted by a range of organisational, technical, legal, and operational factors. These challenges have been widely reported in recent cloud security literature and demonstrate that the complexity of cloud infrastructure extends beyond threat detection alone. Figure 2.1 summarises these challenges into four main categories and provides examples of how each category affects IDS performance, scalability, compliance, and day-to-day security operations within cloud settings. This visual framework supports the discussion that follows by clarifying how different forms of challenges interact and shape the effectiveness of IDS in practice.

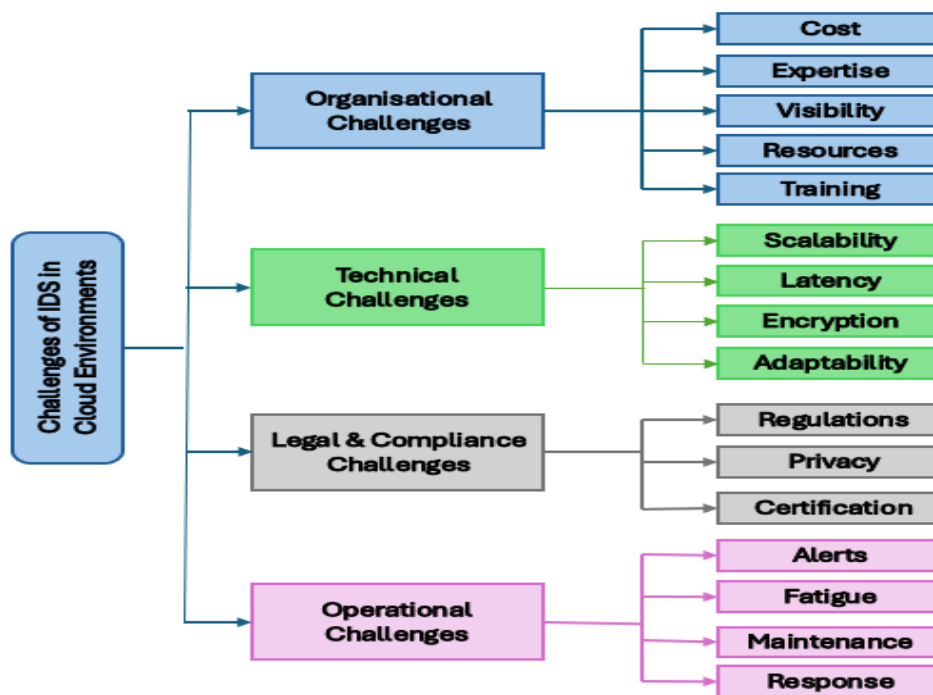


Fig. 2.1 Challenges of IDS in Cloud Environments. Adapted from IDS/IPS in Cloud Environments: Challenges and Solutions, by R. Bhardwaj, 2025, TechTarget360. <https://www.techtarget360.com/ids-ips-in-cloud-environments-challenges-and-solutions/>

Figure 2.1 illustrates the four main categories of challenges — organisational, technical, legal & compliance, and operational — along with specific examples within each category.

2.4.1. Organisational Challenges

As illustrated in Figure 2.1, organisational challenges are linked to cost, visibility, resources, expertise, and training. It is tiring for implementation of lab environments because of different machines, technologies, and the need for consistent security configurations (Almuseelem, 2025). According to Prasad (2024), detection of malicious activities is complicated because many organisations lack complete visibility over their internal network. Results of threats being undetected or reported as false positives is also caused complication that it is difficult to differentiate between malicious and harmless network activities (Alnasser et al., 2025). These issues are made worse by the restricted network visibility, which is difficult because it reduces the ability of signature-based and anomaly-based IDSs to detect sophisticated attacks.

Both organisational and network security issues causes implementation cost of IDS to be high (Almuseelem, 2025). For example, it can be difficult to calculate the return on investment (ROI) for security systems and monitoring ongoing detection systems normally leave organisations with no choice but outsource monitoring to third parties, which increases operational expenses. As companies are left to figure standards themselves due to lack of clear, internationally accepted

guidelines for implementation of IDS in cloud also contribute to compliance challenges, resulting to inconsistent security policies (Almuseelem, 2025).

Teams can effectively implement and manage IDS through skill development and staff training. However, Lata and Singh (2022) emphasised that many organisations face allocation of resource challenges, including computing power to support real-time detection or insufficient storage. Implementation process can be made more difficult by resistance from internal stakeholders, emphasising the importance of organisational support and continuous privacy compliance (Ahmadi, 2024a). This emphasises a trend whereby even organisations with plenty of funds may find it difficult to implement IDS effectively if they lack assistance of experienced staff or internal collaboration.

The cloud implementation cost could be increased by the need to monitor the cloud's wideband data transmission because it requires advanced IDS technologies and experienced staff. Additionally, Riaz et al. (2017) stated that selection process for enterprises is complicated by the lack of standardised metrics for evaluating IDS tools. As cloud security remains one of the main challenges to cloud implementation, Mayuranathan et al., (2022) stated that enterprises without a clear architectural perspective may become confused about who is responsible for data protection and regulatory compliance. Overall, confusion of responsibility, knowledge, and expense to reduce effectiveness of IDS in cloud environments are often connected to issues of organisations.

2.4.2. Technical Challenges

Technical challenges faced by IDS include scalability, latency, encryption, and adaptability, as illustrated in Figure 2.1. Challenges in processing sequential data quickly impact the real-time detection to meet cloud performance requirements and lab implementations for testing IDS are sometimes slow to configure because of variety in systems (Alnasser et al., 2025). Additionally, there's often a difficulty in converting attack traffic into formats of appropriate IDS analysis, especially in test environments with few attack scenarios, which in turn reduces the comprehensiveness of the evaluations (Almuseelem, 2025; Alnasser et al., 2025). This is particularly important in the cloud, where real-world threats are usually more varied than those generated by simulations.

Latency of data processing is one of the main challenges for IDS systems, especially those that include AI integration detection (Almuseelem, 2025). Large-scale log data processing might cause delays that make real-time threat detection impractical if it is not carefully planned for. Moreover, interoperability challenges posed by the variety of cloud architectures; IDS systems need to be flexible enough to support different configurations, but many current solutions still do not have this

capability (Almuseelem, 2025; Prasad, 2024). This results in a trade-off between high-performance detection and wide compatibility.

Traditional Network Intrusion Detection Systems (NIDS) may not be able to handle heavy data effectively and workloads in the cloud, leading to higher rates of false positives and reduced detection accuracy (Ahmadi, 2024a). IDS ability to detect malicious payloads is limited by encryption by preventing them from inspecting traffic contents (Ahmed, 2024; Riaz et al., 2017). Effectiveness of IDS comparison across different cloud providers is the issue because of the lack of standardised metrics for performance, making collaboration more difficult (Ahmadi, 2024a). Regardless of promise, deep learning-based IDS frameworks often lack adaptability, require a lot of resources, which leads to a slow response, and hybrid IDS models tend to be complicated in their optimisation for optimal performance (Alhayan et al., 2025). IDS technical challenges in the cloud go beyond unprocessed detection capability, but also about balancing efficiency, adaptability, and operational sustainability, this is suggested by findings.

2.4.3. Legal and Compliance Challenges

Legal and compliance challenges revolve around regulations, privacy, and certification as illustrated in Figure 2.1. Due to lack of guidelines, this poses a risk to use AI-riven on organisation to process a significant amount of sensitive data (Alhayan et al., 2025). Because of lack of clear regulations, Almuseelem (2025) emphasised that organisations are exposed to risks of mismanagement or privacy obligations.

According to Hassan et al. (2024), major regulations, especially for organisations handling sensitive or personal information, includes Protection of Personal Information Act (POPIA), General Data Protection Regulation (GDPR), and Health Insurance Portability and Accountability Act (HIPAA). This requires not only understanding the legal framework but also ensuring that IDS operations align with these laws without compromising security performance. For example, encryption used for compliance can compromise visibility of IDS into traffic content (Ahmadi, 2024a; Hassan et al., 2024).

Compliance certifications are required for cloud service providers to help solve these issues, which is a key trend (Almuseelem, 2025). Even in the event of certified providers, organisations are still held accountable for data breaches, which lead to confusion in shared responsibility models (Almuseelem, (2025). Additional complexity can be caused by cross-border data transfers as different countries have different security and privacy standards. This suggests that legal issues are more about aligning security policies with an evolving and frequently decentralised regulatory environment than they are about technology detection capabilities.

2.4.4. Operational Challenges

These operational challenges include alerts, fatigue, maintenance, and response as illustrated in Figure 2.1. Human intervention is crucial for validation of alerts because high false positive rates in IDS tools can consume staff time and resources (Almuseelem, 2025). With time, this may result in alert fatigue, where analysts lose attention to security alerts and could fail to recognise real threats (Alnasser et al., 2025). Delayed responses can quickly increase issues in rapid cloud environments; this is caused by this operational pressure.

The complexity to alert management in multi-cloud environment is added by the variance of configurations and services (Zhang et al., 2024). Security teams may be overwhelmed because of ineffective filter and prioritisation mechanisms, reducing overall efficiency for the IDS (Ahmed, 2024). Additionally, machine learning-based IDSs can be very advanced as a result harder for analysts to interpret alerts, especially in “black box” with limited transparency into decision-making processes (Ahmadi, 2024a).

The literature also mentioned the need for balancing training time for IDS models. While too much training increases computational costs and may cause deployment delays, insufficient training increases false positives (Mayuranathan et al., 2022). Signatures need to be updated and adjusted regularly to maintain detection accuracy; but this ongoing maintenance adds to the administrative workload (Shadi Alhakeem & Bin Ajla, 2024). The research shows that organisational workflows, human factors, and the ability to integrate IDS process into larger security operations also cause operational challenges, it is not only poor detection technology.

2.5. Conclusion

This chapter has reviewed literature address to SO1. According to the discussion, cloud computing unique characteristics such as decentralisation, multi-tenancy, and continuous scalability requirements compromise the effectiveness of IDS although it remains crucial security control for detecting and monitoring malicious activities. The need for IDS solutions that adapt in real-time evolving environments is made by these factors because they make detection processes more difficult.

According to the literature, current IDS models, whether hybrid, anomaly-based, or signature-based, struggle to fully address the security requirements of cloud environments. For known threats that has been detected before, signature-based IDS are effective, but they are ineffective against zero-day attacks and require regularly update to their signatures. Anomaly-based IDSs can detect unknown or novel threats, but they suffer from high false positive rates and resource-intensive

operations. Hybrid IDS have limited evidence of successful implementation despite their promising adoption in real-world multi-cloud infrastructures and struggle with scalability and integration issues.

It has been shown that the challenges in implementing IDS in cloud environments go beyond technical ones and include organisational, legal, compliance, and operational challenges. Throughout the literature, major recurring themes include a lack of expertise, latency, cost constraints, encryption complexities, alert fatigue, and privacy concerns. A significant gap still exists in translating these concepts into practical, large-scale solutions that are effective, resource-efficient, and compliant with global regulations, despite numerous experimental and theoretical proposals.

Overall, this review emphasises that solutions that can operate effectively in dynamic cloud environments are needed urgently even though IDS technology has advanced significantly. The absence of standardised datasets, performance benchmarks, and agreed evaluation criteria further limits the ability to compare and validate IDS models, slowing their adoption in real-world cloud deployments. Addressing these gaps will be central to improving IDS capabilities in the cloud and form a foundation for the next phase of this research.

Chapter 3 - Systematic Literature Review

3.1. Introduction

The purpose of this chapter is to presents a Systematic Literature Review (SLR) conducted to address the research question: What are the effective intrusion detection systems and automated threat response mechanisms implemented in cloud environments? The research objective initially started in Chapter 1, which is: To identify effective intrusion detection systems (IDS) and automated threat response mechanisms in cloud environments, directly aligns with this review, designed to address this objective through analysis of relevant academic corpus papers. Chapter 2 reviewed literature on cloud security and intrusion detection to provide theoretical foundation. This chapter address the SO2 by systematically analysing and synthesising prior research to identify current approaches and best practices for improving intrusion detection and threat mitigation by building on these foundations that were addressed in Chapter 2.

A Systematic Literature Review is a research method that is used to collect, analyse, and synthesise existing academic papers relevant to a specific topic, and it is a structured and transparent. Applying the SLR method in this study enables the integration of findings across multiple credible sources and to highlight common challenges, advancements, and research gaps that can inform best practices for cloud-based intrusion detection.

This chapter is organised as follows. Section 3.2 outlines the SLR process, including the databases selected, search strategy, and the inclusion/exclusion rules used to filter the studies. The results and discussion are presented in Section 3.3, where they are analysed systematically across important ideas such threat mitigation approaches, feature selection, and IDS techniques. Section 3.4 concludes the chapter by summarising the insights gained from the review and linking them to the broader objectives of the research.

3.2. Systematic Literature Review Process

Recognised guidelines were followed to ensure transparent, replicable, and methodological rigour. The methodological frameworks commonly used in computer science and information system research guided this review process (Carrera-Rivera et al., 2022; Okoli & Schabram, 2010). The overall goal is to identify relevant studies and synthesise findings in a logical manner to ensure each step is done currently.

The use of PRISMA diagram was the key component for this process. The diagram provides a structured approach for reporting how studies were identified, screened, assessed for eligibility, and

finally included in the review. Transparency is promoted by visually representing the selection process through a PRISMA flow diagram. In this study, the PRISMA-style diagram presented in Figure 3.1 was used to illustrate the number of records identified through database searches, the number of duplicates removed, the studies screened and excluded, and the final studies included for synthesis. The literature selection process was systematic and traceable, reducing potential bias and improving the credibility of the review outcomes; this was ensured by applying PRISMA.

By combining PRISMA guidelines with methodological recommendations from Okoli and Schabram (2010), the SLR process provided a rigorous and recognised guidelines that were followed to ensure transparent, replicable, and methodological rigour. The methodological frameworks commonly used in computer science and information system research guided this review process (Carrera-Rivera et al., 2022; Okoli & Schabram, 2010). The overall goal was to identify relevant studies and synthesise findings in a logical manner to ensure each step is done currently.

Preliminary reading

Before starting the systematic search, there was engagement in preliminary reading to gain understanding on automated threat response and intrusion detection systems (IDS). This enabled the identification of recurrent themes, including hybrid IDS models, the integration of deep learning techniques, as well as the challenges associated with scalability and the occurrence of false positives in cloud-based security. The keywords and search strings were refined through the review of recent studies. The review was not too broad instead it focused and aligned with objective of this chapter; this was acquired through insights from this step.

Research Question

This step provides overview of the research question: What are the effective intrusion detection systems and automated threat response mechanisms implemented in cloud environments?

For this study, this question is important because it addresses SO2. In this step, the real focus is on exploring how different IDS models are implemented in cloud infrastructures to detect potential threats in real-time, including signature-based, anomaly-based, and hybrid systems. It also examines how automated response mechanisms, including machine learning-driven systems, security orchestration, and self-healing frameworks are used to reduce attacks once detected.

PRISMA Diagram

PRISMA flow diagram was used to illustrate how studies were identified, screened, and included in the corpus to ensure transparent and systematic process. The Figure 3.1 summarises the number of

records retrieved from each database, the screening stages applied, and the final number of studies included for analysis.

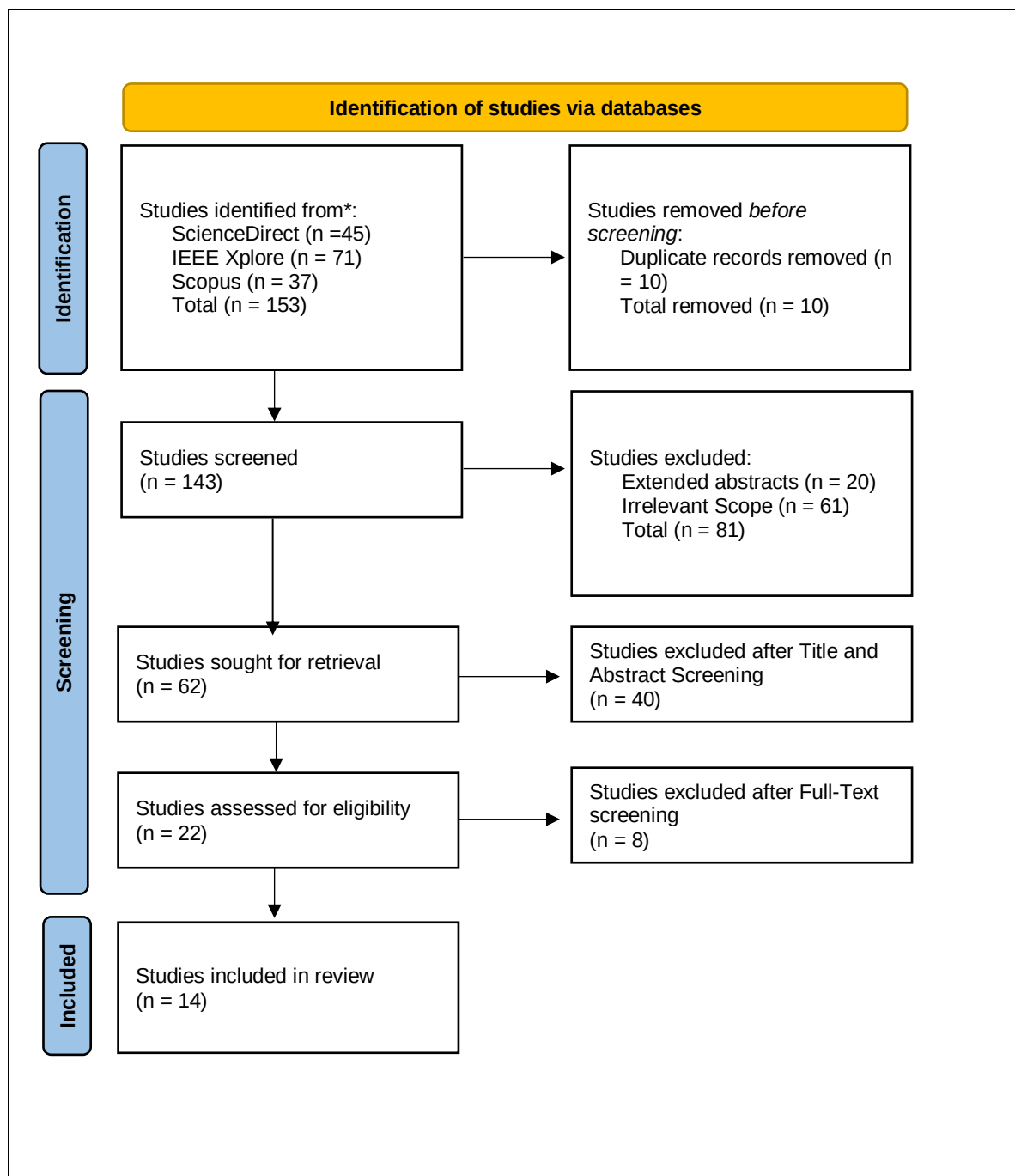


Fig. 3.1. PRISMA-based diagram describing search strategy and outcomes

Protocol

A structured search protocol was followed by this review. ScienceDirect, IEEE Xplore, and Scopus are the databases used because they contain reliable and peer-reviewed studies in information security and cloud computing. To ensure recent and relevant findings, the search targeted publications between 2020 and 2024. Peer-reviewed journal articles, conference papers, and academic were only included publications written in English.

Combination of key terms related to intrusion detection and automated response in cloud settings is the search strategy used. The main search string was:

“(Intrusion Detection Systems” or IDS) AND

“Automated Threat Response” OR “Threat Response Mechanisms” AND

“Cloud Environment” OR “Cloud Security” OR “Cloud Computing”

Search Execution

Three major academic databases (ScienceDirect, IEEE Xplore, and Scopus) were used to carry out search process which was between 26 August 2025 and 5 September 2025. These databases provide high-quality peer-reviewed research in computer science and cloud security, which is why they were selected.

The search was performed on 26 August 2025 for ScienceDirect using keywords ("Intrusion Detection System" OR IDS) AND ("cloud computing" OR "cloud environment") AND ("threat detection" OR "security" OR "attack detection"). Computer Science subject area had limited search to research articles published between 2020 and 2024. Publication titles such as Procedia Computer Science and Journal of King Saud University written in English were only papers included. This search retrieved 45 relevant results.

The search was performed on 30 August 2025 for IEEE Xplore using the keywords ("Intrusion Detection System" OR IDS) AND ("cloud computing" OR "cloud environment") AND ("threat detection" OR "security" OR "attack detection"). The publication types included conferences and journals published between 2020 and 2024, focusing on publication topics related to Cloud Computing. Only open-access publications were included, such as those from IEEE Access. A total of 71 papers were retrieved.

The final search was performed on 5 September 2025 for Scopus using the same keywords ("Intrusion Detection System" OR IDS) AND ("cloud computing" OR "cloud environment") AND ("threat detection" OR "security" OR "attack detection"). The inclusion criteria were limited to publications between 2020 and 2024 within Computer Science subject area, using journal articles and conference papers. Sources included IEEE Access, Advances in Engineering Software, Applied Mathematics and Information Science, Applied Science Switzerland, Big Data Mining and Analytics, and Computers and Security. A total of 71 papers were retrieved.

Further selection

Exclusion Criteria:

During the further selection process after initial retrieval, papers were excluded if they:

- Were duplicates across different databases.
- Were extended abstracts with limited detail.
- Focused only on traditional/on-premises IDS rather than cloud-based IDS.
- Discussed cloud computing security in general terms during screening phase, without focusing specifically on IDS or automated threat response mechanisms.

Further selection process in detail.

After the initial search on ScienceDirect, IEEE Xplore, and Scopus databases, a total of 153 studies were retrieved, 45 from ScienceDirect, 71 from IEEE Xplore, and 37 from Scopus. The first step in the process involved removing duplicate records, which were 10 studies and left 143 studies to be screened.

During the screening phase, total of 81 studies was removed, 20 studies were identified as extended abstracts and did not include enough information for a thorough analysis, and 61 studies were removed due to a lack of relevance to the scope, which did not focus on cloud-based intrusion detection and threat mitigation. This reduced the pool to 65 studies sought for retrieval.

In total, 40 studies were removed for lack of relevance. The remaining studies for full-text assessment were 22. The 8 studies from full-text screening were removed due to a lack of direct relevance and insufficient detail for addressing intrusion detection and automated threat mitigation in cloud environments.

In total, 14 studies were finally selected which met the criteria to serve as the corpus for analysis in this systematic literature review.

In the final review, a total of 14 studies were identified as relevant and included, following the systematic search and screening process described in the previous steps. The base for answering the research question was collectively formed by these papers to analyse and provide evidence: What are the effective intrusion detection systems and automated threat response mechanisms implemented in cloud environments?

Table 3.1 present a list of studies selected for the review, showing each paper’s title, citation, and identification code (P1–P14) used throughout the analysis and discussion.

Table 3.1: Corpus of Papers

Paper ID	Paper Title	Citation
P1	A Deep Learning Approach for Intrusion Detection Systems in Cloud Computing Environments	(Aljuaid & Alshamrani, 2024)
P2	A Literature Review on Attacks Prevention and Profiling in Cloud Computing	(Eddermoug et al., 2023)
P3	A Systematic Literature Review on Cloud Computing Security: Threats and Mitigation Strategies	(Alouffi et al., 2021)
P4	Advanced Persistent Threat attack detection method in cloud computing based on autoencoder and SoftMax regression algorithm	(Abdullayeva, 2021)
P5	An Improved Design for a Cloud Intrusion Detection System Using Hybrid Features Selection Approach with ML Classifier	(Bakro et al., 2023)
P6	Cloud Network Anomaly Detection Using Machine and Deep Learning Techniques - Recent Research Advancements	(Abdallah et al., 2024)
P7	Cloud-Based Intrusion Detection Approach Using Machine Learning Techniques	(Attou et al., 2023)
P8	DeepDefend: A comprehensive framework for DDoS attack detection and prevention in cloud computing	(Ouhssini et al., 2024)
P9	Effective Intrusion Detection System using Hybrid Ensemble Method for Cloud Computing	(Maheswaran et al., 2023)
P10	Enhancing Cloud Security with Deep Learning-Based Intrusion Detection in Cloud Computing Environments	(Devi & Jain, 2024)
P11	Enhancing cybersecurity in cloud computing and WSNs: A hybrid IDS approach	(Sundaramoorthy et al., 2024)
P12	Enhancing intrusion detection: a hybrid machine and deep learning approach	(Sajid et al., 2024)
P13	Machine Learning for Cloud Security: A Systematic Review	(Nassif et al., 2021)
P14	Network Intrusion Detection in Cloud Environments: A Comparative Analysis of Approaches	(Ahmadi, 2024b)
P15		

3.3. Results of Analysis

The analysis goal in this section is to address SO2, which is to identify effective intrusion detection systems (IDS) and automated threat response mechanisms implemented in cloud environments. The selected studies were reviewed and compared according to recurring concepts drawn from the literature, including IDS approaches, ML and DL techniques, feature selection methods, and threats addressed. A concept-centric mapping was developed to organise the findings and highlight how different studies contribute to improving detection accuracy and real-time response. Table 3.2 summarises the presence of these concepts across the 14 reviewed papers.

Table 3.2: Concept-centric map

Concept	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14	
IDS Approaches (Signature/Anomaly)	X	X	X	X	X	X	X	X	X	X	X	X		X	13
ML/DL Techniques Applied	X	X	X	X	X	X	X	X	X	X	X	X	X		13
Feature Selection Methods	X			X	X		X	X	X		X	X			8
Threats Addressed (DDoS, APT, General Intrusions)	X	X	X	X	X	X	X	X	X	X		X	X		12
Threat Mitigation/Response	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14

3.3.1. IDS Approaches (Signature/Anomaly)

In the cloud environments, the primary types of intrusion detection systems that are adopted, include signature-based, anomaly-based, AI-driven, and hybrid systems. Signature-based detection is ineffective against zero-day attacks because it relies on predefined attack pattern but effective against known threats (Ahmadi, 2024a). On the other hand, anomaly-based detection detects deviations from known normal behaviours, which makes it possible to detect new threats but also results in a high rate of false positives (Bakro et al., 2023; Maheswaran et al., 2023).

Hybrid IDS combines strengths of signature-based and anomaly-based techniques to improve accuracy of detection and reduce false positives. Several studies suggest that hybrid models integrate machine learning and deep learning to improve effectiveness in cloud environments (Abdallah et al., 2024; Sajid et al., 2024). For example, for feature classification and extraction, Sajid et al. (2024) presented a hybrid IDS that combine Convolutional Neural Network (CNN) and XGBoost, while Abdallah et al. (2024) employed hybrid clustering and classification with Deep Recurrent Neural Networks. Similarly, Sundaramoorthy et al. (2024) emphasised the importance of hybrid IDS and also developed a hybrid IDS that combines Semi-Supervised Intrusion Recognition (SSIR), One-Class Support Vector Machine (OSVM), Enhanced Long Short-Term Memory (ELSTM), and Multi-Layer Perceptron Neural Network (MLPNN), as well as Wireless Sensor Networks (WSNs).

Deep learning algorithms such as AI-driven, autoencoders, CCNs, and LSTMs are integrated so that solutions can go beyond hybrid approached. To detect and reduce DDoS attacks instantly, Ouhssini et al. (2024) proposed DeepDefend, an anomaly-based IDS that combines deep learning and genetic algorithms. Similarly, to detect Advanced Persistent Threats (APTs) using AI, Abdullayeva (2021) employed autoencoders integrated with SoftMax regression.

The best strategies for cloud environments is the growing trend toward hybrid and AI-driven IDS, which is finally confirmed by literature and systematic literature (Alouffi et al., 2021; Eddermoug et al., 2023). Collectively, even though anomaly-based and signaturred-based are still fundamental, studies show that hybrid and AI-driven systems are the way of the future for addressing the complexity of cloud threats.

3.3.2. ML/DL Techniques Applied

Machine learning (ML) and deep learning (DL) form the cornerstone of modern cloud-based Intrusion Detection Systems (IDS). According Nassif et al. (2021), the high detection accuracy and adaptability of these techniques are leading them to displace traditional methods. Their research identified 30 different ML techniques used in clous security, with Support vector Machines (SVM) being the most common. The fact that ten of these were DL models further signals the increasing importance of neural networks on this domain.

Sajid et al. (2024) combined CNN, XGBoost, and LSTM for network intrusion detection. CNN and XGBoost handled feature extraction, while LSTM handled classification. Similarly, Abdallah et al. (2024) emphasised that DL-based architectures such as LSMTs and Graph Neutral Networks (GNN) are more effective than traditional machine learning techniques such as AdaBoost and Random Forest (RF) when it comes to detection in DDoS traffic.

Specific challenges on a small scale were addressed by developing a specific. To address issues such as high-dimensional data and temporal pattern recognition, Sundaramoorthy et al. (2024) suggested a hybrid IDS framework that combines MLPNN, OSVM, ELSTM, and ISSIR. Devi and Jain (2024) extracted features using a Stacked Contractive Autoencoder (SCAE) and classified using SVM. For anomaly detection in web applications, Maheswaran et al. (2023) tested Autoencoders and Isolation Forests, demonstrating their value in detecting traffic variations.

Integrating DL techniques for DDoS detection has been the subject for several studies. Ouhssini et al. (2024) developed DeepDefend which uses CNN-LSTM-Transformer networks to detect in real-time. Additionally, their framework combines CNN and LSTM layers with decision trees and random forests, emphasising the adaptability of hybridised DL techniques. Similarly, Aljuaid and Alshamrani

(2024) implemented CNN architectures to achieve high detection performance, achieving over 98% accuracy.

Traditional ML algorithms ongoing value in cloud IDS is further supported by other studies. DL provide higher accuracy but higher processing requirements. The trade-offs between ML and DL models is highlighted by Bakro et al. (2023). Attou et al. (2023) applied RF, SVM, KNN, and Naïve Bayes to classification tasks. Abdullayeva (2021) employed autoencoder-based neural networks with a SoftMax regression layer for classification to detect APT. To identify advanced threats such as SlowDoS attacks in encrypted traffic, Alouffi et al. (2021) explored clustering when combined with DL.

These findings are supported by review studies, which demonstrate that DL models are crucial for managing zero-day and complex threats, while ML algorithms are best suited for detecting known attacks (Eddermoug et al., 2023). The literature confirms whether standalone or hybrid, ML/DL approaches are crucial for enhancing IDS effectiveness in cloud environments. CNN, RF, LSTM, and autoencoder models are frequently mentioned as the most influential models.

3.3.3. Feature Selection Methods

Feature selection is a crucial step for enhancing the effectiveness of cloud-based IDS, as it improves classification accuracy while reducing data dimensionality. This importance is widely supported in the literature, with studies confirming that selecting the most relevant features directly improves detection rates and reduces computational overhead (Abdullayeva, 2021; Bakro et al., 2023).

To enhance feature selection, researchers have widely explored hybrid approaches. To eliminate irrelevant characteristics and focus on their model on the most crucial traffic patterns for intrusion detection, Sajid et al. (2024) integrated CNN and XGBoost. Similarly, Sundaramoorthy et al. (2024) utilised an OSVM and ISSIR in their IDS framework. ISSR determined the influence of features to help to improve the datasets. Genetic algorithms have also become more popular in this space. Ouhssini et al. (2024) employed them to find feature subsets that improved the accuracy of real-time DDoS detection.

Researchers are increasingly combining meta-heuristics with traditional statistical methods to enhance feature selection. For example, Bakro et al. (2023) created a hybrid framework that combine Information Gain and Chi-Square tests with Particle Swarm Optimisation (PSO), generating more robust and accurate feature subsets. Similarly, Abdullayeva (2021) demonstrated that deep learning-based autoencoders excel at this task by automatically learning to reduce data

dimensionality without human intervention, all while preserving the essential patterns needed for effective intrusion detection.

More recent work also integrates sophisticated correlation with visualisation techniques to refine feature sets. Aljuaid and Alshamrani (2024) employed Binary Particle Swarm Optimisation (BPSO) and Pearson correlation to identify highly correlated features, which enhances model accuracy and reduce redundancy. Collectively, feature selection is a crucial part of creating IDS frameworks is being demonstrated by these techniques, these frameworks can manage the complexity and scale of cloud environments instead of being a side project.

3.3.4. Threats Addressed (DDoS, APT, General Intrusions)

Vulnerability of cloud environments can be compromised by cyberthreats, including Advanced Persistent Threats, general network intrusions, and Distributed Denial of Service (DDoS). Most reviewed studies discuss DDoS attacks because they can quickly overwhelm cloud resources and disrupt availability. For example, Abdallah et al. (2024) and Nassif et al. (2021) highlighted that for maintaining reliability of service in cloud-based systems, DDoS is a major concern. Frameworks such as DeepDefend demonstrated significance of this threat in cloud IDS research, which are made especially for DDoS detection and prevention (Ouhssini et al., 2024).

APT attacks pose a serious risk because of their long-term and hidden nature despite being less discussed about. Abdullayeva (2021) addresses this gap by proposing autoencoder-based deep learning technique for detecting APTs in cloud environments and emphasising how traditional IDS models often fail to detect such sophisticated threats. This study emphasises the need for specific detection methods as APTs are increasingly being acknowledged as separate category.

General intrusions such as port scans, botnets, brute force, and unauthorised access are generally acknowledged by IDS studies. While Maheswaran et al. (2023) and Sajid et al. (2024) focus on multi-class intrusion detection systems that can detect both novel and known threats, Aljuaid and Alshamrani (2024) and Eddermoug et al. (2023) emphasised hybrid IDS methods that can detect a wide range of attacks such as remote-to-user and user-to-root exploits.

Collectively, more attention is being drawn to addressing more sophisticated threats such as APTs and a wide range of general intrusions although the literature demonstrates that DDoS attacks are the focus of current IDS research (Alouffi et al., 2021). To prevent immediate disruptions, there is a need to create IDS frameworks that protect cloud environments from large scale, hidden, and evolving attacks, this is demonstrated by the literature.

3.3.5. Threat Mitigation/Response

The basis is provided by traditional network intrusion detection systems (NIDS), which maintain traffic, detect vulnerabilities, and ensure compliance to regulations such as GDPR and HIPAA (Ahmadi, 2024a). The integration of deep learning and machine learning approaches for more effective response systems is led by limitation to address zero-day threats (Nassif et al., 2021).

When an attack is detected, automated response modules in hybrid and AI-driven IDS frameworks often alert administrators or take immediate action. Sundaramoorthy et al. (2024) emphasised real-time responses through MLPNN, strengthening defences against evolving threats. Sajid et al. (2024) demonstrated how hybrid IDS models can enhance detection rates while reducing false positives. Devi and Jain (2024) determined the priority of mitigation strategies in which the type of attack detected, emphasising the significance of policy-driven intrusion response.

Mitigation extends beyond simple detection with advanced frameworks. Ouhssini et al. (2024) enabled proactive mitigation instead of post-attack analysis to enable real-time response to DDoS attacks. Abdullayeva (2021) proposed an APT mitigation technique that combines two-factor authentication and deep learning autoencoders to improve cloud defences. Similarly, Aljuaid and Alshamrani (2024) demonstrated that CNN-based IDS enable immediate responses to reduce cloud security threats by classifying real-time intrusion.

Alouffi et al. (2021) suggest the Autonomous Cloud Intrusion Response System (ACIRS) for utility-based attack response, while Abdallah et al. (2024) argue that integrating IDS with firewalls and continuous monitoring improves resilience against DDoS. Adaptive reconfiguration and intrusion prevention systems (IPS) are crucial components of effective mitigation (Eddermoug et al., 2023).

In summary, AI-enhanced and policy-driven responses mechanisms has evolved to active monitoring as the primary method of cloud threat mitigation. Addressing new evolving threats such as APT and DDoS requires integration of DL/ML models, layered defence systems, and real-time response frameworks because traditional IDS provide basic level of protection. According to the literature, future of cloud threat mitigation is demonstrating the adaptive, proactive, and automated response systems.

3.4. Conclusion

This chapter presented a systematic literature review (SLR) that aimed to identify effective intrusion detection systems (IDS) and automated threat response mechanisms implemented in cloud environments. Using a systematic review process, relevant studies were gathered, screened, and assessed for eligibility. This process qualified them to be reviewed. The review narrowed down from initial 153 studies to 14 high quality studies that addressed the research question.

The findings emphasise that the cloud security still face an evolving challenge, this includes problems with intrusion detection and DDoS attacks mitigation. The effectiveness of deep learning and machine learning is analysed and demonstrated by studies, integrating detection systems with cooperative frameworks to enhance detection accuracy and reduce false positives. Additionally, an increasing trend towards response systems that are automated and adaptive that detect in real-time is showed by the literature, reflecting the dynamic nature of cloud environments.

The SLR not only provides evidence for current solutions, but it also identifies gaps in current research. For example, while primarily focus on many research studies on cloud is detection, fewer studies provide scalable, automated response defences to reduce emerging complexity. The identified gap indicates the need for future research to move beyond passive detection. This indicate that future studies to focus on integrating IDS with automated and proactive-response mechanisms, creating a unified defensive system that cannot only identify threats but also contain and respond to them effectively.

In conclusion, the SLR enhances the objective of this research by integrating multiple approaches, understanding their strengths and limitations, and emphasising future work. It helps in gaining a strong foundation in the next stages of this study while still linked to the broader objective of developing best practices for intrusion detection and threat mitigation in cloud environments.

Chapter 4 - Research Design

4.1. Introduction

The purpose of this chapter is to outline the research design and methodology that guide this study. The primary research objective is to develop best practices for enhancing intrusion detection and threat mitigation strategies in cloud environments as stated in Chapter 1. The research design provides a structured plan that links the stated objectives to achieve this, with the methods used to collect and analyse primary data.

This chapter is organised as follows. Section 4.2 presents the overall research process and links each objective to its corresponding method. Section 4.3 explains the primary data collection strategy, while Section 4.4 discusses participant sampling. Section 4.5 outlines the instrument design. Section 4.6 addresses ethical considerations. Finally, Section 4.7 provides a conclusion by summarising the methodological approach and its relevance to the study.

4.2. Research Process

The research process for this study is summarised in Figure 4.1. The diagram visually links the research problem (slow adaptability of intrusion detection and threat mitigation in cloud environments) to the Primary Objective (PRO) and to the three Secondary Objectives (SO1–SO3). It also shows how each objective is aligned to a specific method and to the envisaged output for that stage of the investigation. The left-to-right flow illustrates how the study moves from identifying challenges through literature review, to identifying effective systems through systematic literature review, and finally to obtaining practical, practice-based insights through semi-structured interviews. The final stage which is critical reasoning that integrates these outputs into a consolidated set of best practices aimed to address the PRO.

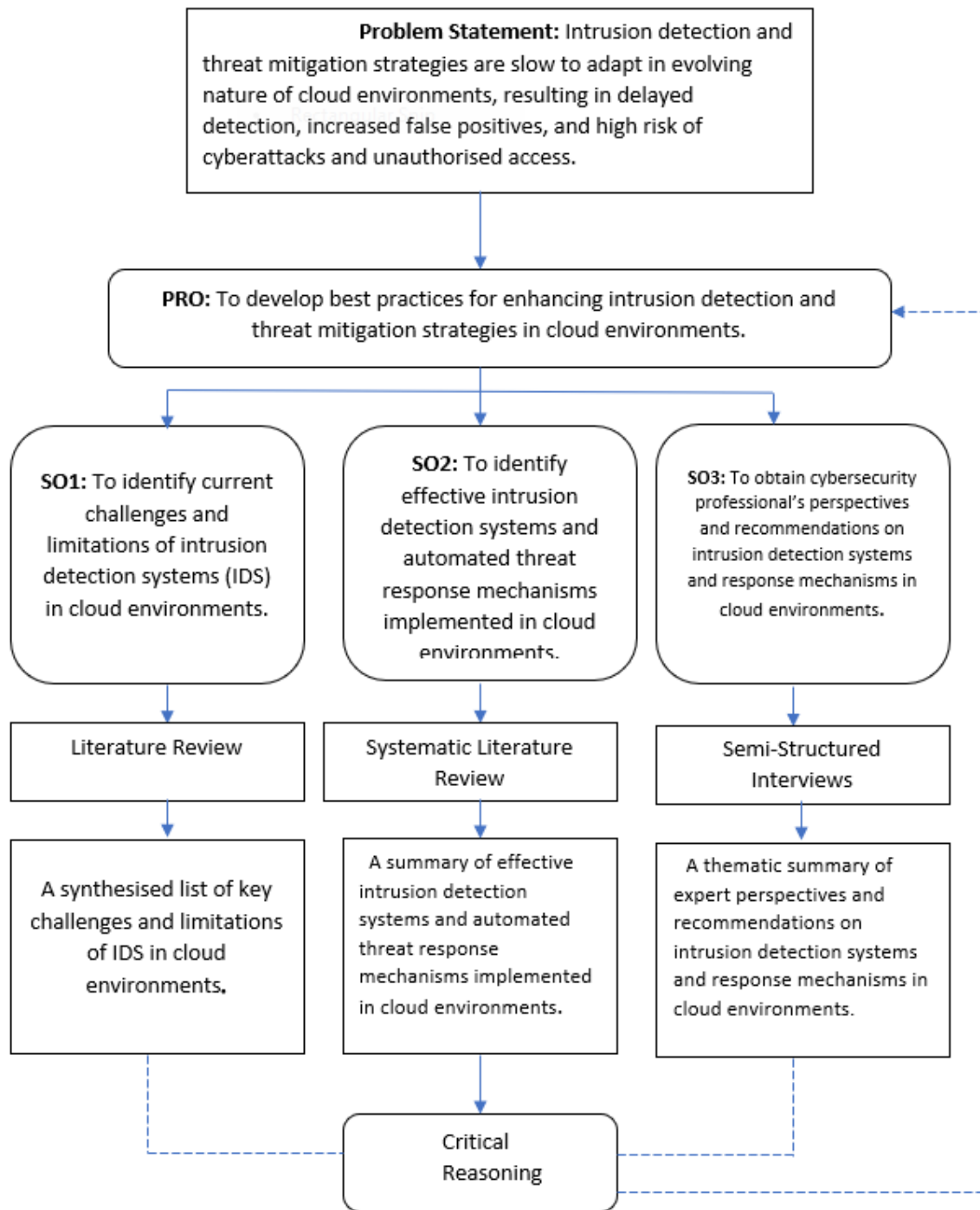


Figure 4.1: Research Process Diagram

For this study, to address the research problem presented in Figure 4.1, which was initially stated in Chapter 1, research process is structured, emphasising the slow adaptability of intrusion detection and threat mitigation strategies in cloud environments. This problem results in delayed detection, increased false positives and high risks of cyberattacks and unauthorised access. The research process is therefore designed to systematically address this problem by combining both secondary and primary data collection methods to ensure a comprehensive response to the problem.

The research process adopted in this study is designed to produce outputs that collectively contribute to the achievement of the Primary Objective (PRO). This process ensures that each

objective is addressed through a methodologically approach. As illustrated in Figure 4.1, each research objective has a specific related method and envisaged output. Before integrating these findings into a set of documented best practices, this structured process moves systematically from the identification of challenges to the evaluation of effective mechanisms, and finally to the collection of expert insights.

A qualitative method is used for this study, as initially stated in Chapter 1. The PRO of the study is to develop best practices for enhancing intrusion detection and threat mitigation strategies in cloud environments. The study applied critical reasoning to achieve this objective, which according to Potter (2012), involves the logical synthesis of findings to reach justified conclusions. To integrate the outputs of SO1, SO2, and SO3 into a consolidated best practice, critical reasoning was applied to this study.

Literature review allows exploration of academic and industry sources to identify the current challenges and limitations of intrusion detection systems in cloud environments, addressing SO1. As Liang and Xu (2025) and Snyder (2019) emphasised, a literature review synthesises existing knowledge to establish what is already known and where gaps exist. The literature review provides an envisaged output in this study: a synthesised list of IDS challenges and limitations that must be considered in developing best practices as discussed in more detail in Chapter 2 of this study.

Systematic literature review applies a replicable and structured process to identify, evaluate, and synthesise relevant studies, addressing SO2. By using specific inclusion and exclusion criteria, according to De Souza et al. (2022) and Van Dinter et al. (2021), SLRs are rigorous because they reduce bias. In this study, the SLR focused on identifying effective IDS and automated threat response mechanisms in cloud environments. The envisaged output, which is discussed in more detail in Chapter 3 of this study, is a detailed summary of mechanisms that have been validated in academic and industry research.

SO3 is addressed using semi-structured interviews, which were selected because they allow conversation to develop naturally while still being guided by predefined questions. This format gives participants the ability to explain their experiences and perspectives in detail, making it suitable for topics in cybersecurity where practical insights are not fully captured (Adeoye-Olatunde & Olenik, 2021). In this study, interviews will be conducted with cybersecurity professionals who work directly with intrusion detection and automated response mechanisms in cloud environments. Their contributions provide real-world perspectives on operational challenges and effective practices. The findings from these discussions will be organised into thematic categories, enabling comparison with trends identified in the literature.

Together, these methods ensures triangulation, enhance reliability and validity of the findings (Johnston et al., 2021). The development of a validated set of best practices is led by integration of findings through critical reasoning, thereby achieving the research objective.

4.3. Primary Data Collection

The purpose of the primary data collection is to obtain expert insights from cybersecurity professionals on the effectiveness, challenges, and future directions of intrusion detection systems (IDS) and automated threat response in cloud environments. This supports SO3 by complementing findings from the literature with practical, real-world perspectives.

Through open-ended questions, semi-structured interviews will be used as the primary data collection method. An interview guide in Appendix A, has been prepared, informed by the gaps and themes identified in the literature review and systematic literature review. Flexibility for participants to elaborate on their experiences will be allowed by interviews, ensuring depth of all key themes relevant to the research.

To accommodate availability of the participants and ensure flexibility across different locations, interviews will take place online (via Microsoft Teams).

Data collection will start after obtaining REC-H approval (obtaining research ethics approval from the university).

4.4. Participant Sampling

Cybersecurity professionals will be the participants for this study because of their direct experience in cloud security, IDS, and automated threat response. These individuals to provide insights into the practical challenges and solutions associated with intrusion detection in cloud environments, making them most suitable. The findings of the literature will be complemented by applied, practice-based knowledge through cybersecurity professionals' perspectives and recommendations.

A sample is a smaller group selected from the population. A combination of convenience and purposive sampling will used by this study. An element of convenience sampling will be used to allow completion of the interviews. As emphasised by Stratton (2021), convenience sampling refers to selecting participants who can be easily accessible to researcher. By conducting interviews online (via Microsoft Teams), it will be easy to meet the participants regardless of their location or time constraints by conducting interviews online, via Microsoft Teams. This approach increases the likelihood of successful participation and timely completion of data collection, while still maintaining a focus on relevant expertise.

Additionally, purposive sampling will be applied in this study to select the participants that meet the relevant criteria to the research objective. According to Ahmad and Wilkins (2025), purposive sampling is crucial in qualitative research because it selects individuals with specialised knowledge. For this study, the criteria include at least two years of experience in cybersecurity, direct involvement in cloud security or IDS practices, and willingness to participate in an interview of approximately 30-45 minutes.

Between five and eight participants is what the study aims to recruit. Rather than using statistical representatives, data saturation will be used to determine sample size. According to O'Brien et al. (2014), data saturation aligns the range with the standards of qualitative research. Saturation refers to the point at which no new themes, insights, or patterns emerge from the data, making further data collection unnecessary because additional responses would likely be repetitive rather than contributory (Saunders et al., 2018). The study focuses on saturation by prioritising the depth and richness of information rather than the number of participants. Such a sample is expected to provide a sufficiently diverse set of perspectives, while also allowing for the feasibility and manageability of the data collection.

4.5. Instrument Design

A semi-structured interview guide will be used to help address the objectives of the study. Each question is informed by ideas drawn from the literature, but the wording and format remain open enough for participants to explain their views in their own way. Following the structure recommended by Adeoye-Olatunde and Olenik (2021), the questions are grouped into four sections: Demographics, Challenges and Limitations (SO1), Solutions and Best Practices (SO2), and Professional Perspectives and Recommendations (SO3).

Section A: Demographics

Questions in Table 4.1 are used to interpret participants responses within the scope of this study by establishing participants background and context. As emphasised by Adeoye-Olatunde and Olenik (2021) demographic information provides important context in qualitative research to ensure that findings can be accurately situated within professional experiences. This section allows for the contextualisation of responses based on participants' professional experience, organisational scale, and exposure to cloud security environments.

Table 4.1: Demographic Questions

Section A	Question
A1	What is your current role in the field of cybersecurity?
A2	How many years of experience do you have in cloud security?
A3	What type of organisation do you work in?
A4	Can you describe scale of your organisation's cloud adoption?

Section B: Challenges and Limitations

Questions in Table 4.2 are used to explore challenges and limitations of IDS in cloud environments. Chapter 2 explained in detail issues such as insufficient datasets, encrypted data challenges, poor performance under high traffic, and cloud-specific vulnerabilities (Riaz et al., 2017; Lata & Singh, 2022). This section aligns with SO1, focusing on identifying the main challenges that affect IDS operation in cloud environments.

Table 4.2: IDS Challenges and Limitations Questions

Statement	Source	Question
IDS often fail in high-traffic environments, making detection unreliable.	(Riaz et al., 2017)	B1. In your experience, what challenges have you observed with IDS performance under high-traffic conditions?
Lack of datasets is a significant barrier to evaluating IDS in cloud.	(Lata & Singh, 2022b)	B2. How would you describe the availability of standard datasets for evaluating IDS in cloud environments?
Handling encrypted data remains a critical challenge.	(Alouffi et al., 2021)	B3. In your perspective, what difficulties do IDS face when handling encrypted data in cloud environments?
Securing cloud-specific components such as hypervisors, SDN, and VM migration remains difficult.	(Riaz et al., 2017)	B4. What issues have you encountered (if any) with securing cloud-specific components such as hypervisors, SDN, or VM migration?

Section C: Solutions and Best Practices

To capture participants insights on potential mechanisms and improvements for IDS in cloud environments, questions in Table 4.3 are used. According to literature proposed in Chapter 3, automated response systems, IDS-as-a-Service, and ML/DL-based IDS are all promising solutions (Abdallah et al., 2024; Sajid et al., 2024). SO2 aim to identify effective IDS techniques, aligning with this section. By referencing those key studies on automation and intelligent IDS, there can be a good balance between theory and practice. This allows for the judgement not just the potential of these new technologies, but how realistic and practical they would be to implement.

Table 4.3: IDS Solutions and Best Practices Questions

Statement	Source	Question
ML/DL techniques are widely used to improve IDS accuracy.	(Abdallah et al., 2024; Sajid et al., 2024)	C1. What are your views on the effectiveness of ML or DL-based approaches in improving IDS accuracy in cloud environments?
IDS-as-a-Service frameworks are emerging for continuous monitoring.	(Alouffi et al., 2021)	C2. How feasible is the adoption of IDS-as-a-Service (IDSaaS) for organisations using cloud platforms?
Automated response systems can reduce delays in mitigation.	(De Souza et al., 2022b)	C3. What role do you see automated response systems playing in mitigating threats in real time?
Architectural improvements remain essential for IDS advancement.	(Snyder, 2019b)	C4. In your opinion, which design, or architectural improvements are most critical for improving IDS in cloud environments?

Section D: Professional Perspectives and Recommendations

Questions in Table 4.4 are used to capture expert insights on applied strategies, including risk assessment, sector-specific practices, Zero Trust, and CSP features. Hasan (2024) and Hassan et al. (2024) emphasised that these strategies reflect practical considerations beyond purely technical mechanisms. SO3 aims to obtain cybersecurity professionals perspectives and recommendations, aligning with this section. Across industries, it bridges the gap between theoretical best practices and practical implementation strategies. Effectiveness of IDS deployment is strengthened by integrating contextual factors such as organisational policies, risk tolerance, and regulatory compliance.

Table 4.4: Professionals Perspectives and Recommendations

Statement	Source	Question
Risk assessment is critical in implementing cloud-based IDS.	(Hassan et al., 2024)	D1. How important is risk assessment and management when implementing IDS in cloud environments?
CSP features such as IAM, and encryption can strengthen IDS.	(Alouffi et al., 2021)	D2. To what extent does your organisation rely on security features provided by cloud service providers?
Zero Trust Architecture reduces insider threats by 35–40%.	(Hasan, 2024)	D3. What is your experience on implementing Zero Trust Architecture (ZTA) in cloud environments? Have you observed improvements in reducing insider threats?
Sector-specific challenges require tailored best practices.	(Johnston et al., 2021)	D4. What sector-specific challenges or considerations do you think are most significant when implementing IDS in cloud environments?

4.6. Ethical Considerations

This study will comply with Nelson Mandela University Research Ethics Committee (REC-H) guidelines and ethical risk assessment framework (Appendix B). According to REC-H guidelines, all research involving human participants must undergo prior ethics review, with the level of review determined by the risk classification of the project (low, medium, or high). Risk is defined as a function of both the probability of harm occurring and the magnitude of that harm.

This study is considered as low-risk research according to REC-H assessment form in Appendix B. Cybersecurity professionals are the participants, who are not regarded as vulnerable group, and the topic is technical nature, focusing on intrusion detection systems and response mechanisms in cloud environments. Therefore, no sensitive personal data will be collected. Minor inconvenience with regards to time required for the interviews is the only foreseeable risk to participants, which falls under REC-H's definition of low risk ("where the only foreseeable risk is one of inconvenience or discomfort").

Several safeguards will be implemented throughout the study to ensure compliance with ethical standards. The purpose of the study, its objectives, and what their participation entails will be clearly classified by giving each participant an information sheet to ensure informed consent is obtained in

all participants. Written consent will be secured before any interviews are conducted to ensure that all participants are fully aware of their participation.

Individuals' participation will remain voluntary, so they will have a right to withdraw from the study at any time without facing any consequences. This safeguard protects participants freedom of choice throughout the research process.

Confidentiality and anonymity will also be strictly maintained. During both transcription and reporting, responses will remain anonymous, and no personal or organisational identifiers will be disclosed in the final report. This measure ensures that participants' professional insights are protected and cannot be traced back to them directly.

Additionally, data security will be prioritised by storing all digital files on password-protected devices. The collected data will be securely stored for the required institutional retention period before being permanently destroyed in accordance with Nelson Mandela University's research policy.

Finally, the study has been designed to reduce any risk of harm. There is no exposure to physical, psychological, social, or legal harm as the focus is limited to professional perspectives. This makes the research low risk.

4.7. Conclusion

This chapter outlined the adopted methodological framework to address the research problem presented in Figure 4.1. This study used a qualitative research design to explore the challenges and limitations of intrusion detection systems in cloud environments, evaluate best practices and mechanisms identified in existing literature, and obtain perspectives and recommendations from cybersecurity professionals. The research process was structured to align each secondary and primary objective with appropriate methods and envisaged outputs, ensuring that the Primary Research Objective (PRO) could be addressed systematically.

To ensure that the questions posed were not only relevant but also capable of probing insights directly linked to the research objectives, both theoretical foundations and findings from prior studies informed design of the research instruments and the interview schedule. Ethical considerations were in accordance with Nelson Mandela University REC-H guidelines, being carefully integrated into the research design, with specific attention given to issues of informed consent, confidentiality, voluntary participation, and data security.

The proposed methodological framework demonstrates how the study could be implemented to generate valid and reliable findings although primary data collection will not be conducted in practice. This provides a clear plan for operationalising the research should data collection be

pursued in the future. The next chapter will build upon this framework by presenting the analysis and interpretation of findings based on the structured research process.

Chapter 5 - Conclusion

5.1. Introduction

The purpose of this chapter is to present the overall conclusion of the study, summarises key findings presented in the previous chapters, and emphasise how the research objectives have been achieved. Research problem, which stated the challenge of traditional security approaches in adapting to evolving cloud environments, was introduced in Chapter 1. Primary objective of this study – to develop best practices for enhancing intrusion detection and threat mitigation strategies in cloud environments, was also introduced in Chapter 1. The three secondary objectives were addressed in chapters 2, 3, and 4 by identifying current challenges in cloud through literature review (SO1), evaluating effective mechanisms through a systematic literature review (SO2), and obtaining professional perspectives and recommendations through semi-structured interviews (SO3).

This chapter show how the PRO has been achieved by combining the findings of these chapters. The chapter discusses the limitations, potential contribution, and future directions. Reflections on the overall research process and lessons learned are included at the end.

The chapter is structured as follows. Section 5.2 discusses how each objective was met. Section 5.3 emphasises the potential contribution of this study. Section 5.4 outlines the limitations of the study. Section 5.5 suggests areas for future research. Section 5.6 concludes with personal reflections on the research process.

5.2. Meeting the Research Objectives

To develop best practices for enhancing intrusion detection and threat mitigation strategies in cloud environments is the primary objective of this study. Each secondary objective was addressed through a qualitative analysis towards the progress of primary research objective.

SO1: To identify current challenges and limitations of intrusion detection systems (IDS) in cloud environments.

This objective was fully achieved in Chapter 2 through a literature review. It explored various academic sources and industry reports. Key challenges were highlighted by the findings, including cost limitations, lack of expertise, latency, encryption issues, privacy concerns, and alert fatigue to dynamic cloud settings. These insights provided foundation for identifying improvement opportunities and a clear understanding of the existing weakness in IDS implementation.

SO2: To identify effective intrusion detection systems and automated threat response mechanisms implemented in cloud environments.

This objective was also fully achieved in Chapter 3 through a systematic literature review. The best practices such as the integration of machine learning models, hybrid detection approaches integrating signature and anomaly-based strategies, and the use centralised monitoring tools to improve visibility across multi-cloud environments were revealed by analysis. Additionally, automation and continuous updates were identified as essential for maintaining IDS effectiveness against evolving threats.

SO3: To obtain cybersecurity professionals perspectives and recommendations on intrusion detection systems and response mechanisms in cloud environments.

This objective was not achieved as it falls outside the scope of Postgraduate Diploma report. The actual primary data collection will be undertaken in a future phase beyond current scope of the study although semi-structured interview framework and guiding questions were developed in Appendix A.

PRO: To develop best practices for enhancing intrusion detection and threat mitigation strategies in cloud environments.

This objective has been partially achieved through the analysis of existing literature. The primary data collection and analysis will be completed in future stages of the research, with set of preliminary best practices being outlined, which will be further refined after the completion of primary data.

5.3. Potential Contribution of this Study

This study could help expand the literature regarding cloud security by identifying and evaluating best practices for intrusion detection and threat mitigation. It systematically shows how techniques of intrusion detection can be adapted to address the unique challenges of distributed and dynamic cloud environments.

The problem statement outlined in Chapter 1 directly links with the research, which emphasised that intrusion detection and threat mitigation strategies are slow to adapt in evolving nature of cloud environments, resulting in delayed detection, increased false positives, and high risk of cyberattacks and unauthorised access. The study addressed this problem through a systematic literature review by evaluating evidence-based practices that improve detection accuracy, improve overall cloud infrastructure resilience, and reduce false positives.

The findings of this research present a potential theoretical and practical contribution although the primary objective has only been partially achieved due to the absence of primary data collection. Future research and practical implementations can build upon solid conceptual foundation that is established by a study. Researchers and cybersecurity professionals can develop adaptive security frameworks such as automation, intelligent threat analytics, and continuous monitoring can be developed by researchers, cybersecurity professionals, and organisations using guide gained from these insights.

This study provides the foundation for potentially achieving the primary research objective by defining best practices and theoretical principles that future empirical studies can build upon to develop a comprehensive intrusion detection and threat mitigation framework for cloud environments.

5.4. Limitations of Study

Time constraints and the scope of the postgraduate diploma research are the primarily related limitations that this study faced. The systematic literature review included only 14 studies due to the limited timeframe, which may not fully capture the entire range of existing research on intrusion detection and automated threat response in cloud environments. Despite being structured and methodologically sound, the review was not as extensive or detailed as it was initially intended.

Additionally, the primary data collection objective was not achieved, as it falls outside the scope of this postgraduate diploma research report. The duration of this study was not feasible because conducting empirical research would have required additional time for ethical approval and data collection procedures. As a result, there is limited ability to validate results through practical implementation or professional perspectives because findings are based solely on secondary data.

The study still provides meaningful insights by consolidating existing knowledge and identifying research gaps despite these constraints. Future research aimed at fulfilling the primary objective through empirical validation and framework development can be informed by these findings.

5.5. Future Research

Future research must therefore address the limitations discussed in this study through primary data collection and empirical validation. More specifically, the next stage is to develop and test an intrusion detection and threat mitigation framework specifically designed for cloud environments.

Such research would include the collection of data from cybersecurity professionals and tested on cloud service providers to evaluate the accuracy, effectiveness, and scalability of real-world

conditions. Additionally, automation and adaptive learning techniques such as AI -driven detection and security orchestration tools models must be incorporated to explore efforts to reduce high false positives and improve response time.

The main objective of future studies will be to develop best practices for enhancing intrusion detection and threat mitigation strategies in cloud environments. Such research would contribute practical insights and provide measurable outcomes that complement the theoretical foundation established in this report.

5.6. Reflections

For this study, research process has been both challenging and rewarding. Conducting a systematic literature review required careful planning, persistence, and critical analysis of existing academic sources. Databases such as ScienceDirect, IEEE Xplore, and Scopus were used extensively to retrieve relevant papers. Referencing tool such as Zotero assisted with citation and referencing management and maintaining academic integrity, while Microsoft excel was used to organise and filter studies according to inclusion and exclusion criteria. Throughout this process, there was development of a deeper appreciation for the rigour required in academic research and the importance of following a structured and transparent methodology.

One of the key lessons learned was the value of precision when defining research objectives and questions. A well-formulated research question provided clear direction and made it easier to identify relevant studies while avoiding unnecessary material. To critically evaluate what studies were reporting, specifically their limitations, validity, and context is another lesson learned. This ability has ensured better synthesising information from multiple sources and presenting it logically.

Given the depth of analysis that systematic literature review required, time constraints were a major limitation. With a need to find a balance between academic demands and other coursework, the body of literature needed to be narrowed down to 14 core studies. Additionally, navigating research ethics processes and understanding the distinction between secondary and primary research required extra effort, especially since primary data collection was outside the scope of this postgraduate diploma.

Future students are advised to start their research early and to maintain consistent progress throughout the year. To seek feedback regularly from mentors and lecturers is also essential as this helps to refine both structure and content. Final advise is that students should not just view research only as a requirement for completion but a process, but also the opportunity to develop lifelong problem-solving and analytical skills that are important in both academic and professional settings.

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Appendix A – Research Instrument

Interview Guide

Opening Script

Good day, thank you for agreeing to participate in this interview. My name is Banele, and I am conducting this research as part of my postgraduate studies in Cybersecurity at Nelson Mandela University.

The purpose of this study is to explore best practices for enhancing intrusion detection and threat mitigation in cloud environments.

This interview will take about 30–45 minutes. Your participation is voluntary, and you may withdraw at any time without penalty. All responses will be kept confidential and used for academic purposes only. Do I have your consent to proceed and to record this interview for transcription purposes?

Interview Sections

Section A: Demographics

- **A1.** What is your current role in the field of cybersecurity?
- **A2.** How many years of experience do you have in cloud security?
- **A3.** What type of organisation do you work in?
- **A4.** Can you describe the scale of your organisation's cloud adoption?

Section B: Challenges and Limitations

- **B1.** In your experience, what challenges have you observed with IDS performance under high-traffic conditions?
- **B2.** How would you describe the availability of standard datasets for evaluating IDS in cloud environments?
- **B3.** In your perspective, what difficulties do IDS face when handling encrypted data in cloud environments?
- **B4.** What issues have you encountered (if any) with securing cloud-specific components such as hypervisors, SDN, or VM migration?

Possible follow-up question: Can you give an example of a situation where this issue significantly affected detection accuracy?

Section C: Solutions and Best Practices

- **C1.** What are your views on the effectiveness of ML or DL-based approaches in improving IDS accuracy in cloud environments?
- **C2.** How feasible is the adoption of IDS-as-a-Service (IDSaaS) for organisations using cloud platforms?
- **C3.** What role do you see automated response systems playing in mitigating threats in real time?

- **C4.** In your opinion, which design, or architectural improvements are most critical for improving IDS in cloud environments?

Possible follow-up questions: What challenges have you observed when implementing these approaches?

Have you encountered instance where automation failed in mitigating a threat?

Section D: Professional Perspectives and Recommendations

- **D1.** How important is risk assessment and management when implementing IDS in cloud environments?
- **D2.** To what extent does your organisation rely on security features provided by cloud service providers?
- **D3.** What is your experience on implementing Zero Trust Architecture (ZTA) in cloud environments? Have you observed improvements in reducing insider threats?
- **D4.** What sector-specific challenges or considerations do you think are most significant when implementing IDS in cloud environments?

Possible follow-up question: What were the main challenges during the implementation phase?

Closing Script

Thank you very much for your time and insights. Your input is extremely valuable to this study and will help in developing practical best practices for improving intrusion detection in cloud environments.

Before we end, is there anything else you would like to add that we may not have covered?

If you would like, I can share a summary of the findings once the study is completed. Thank you again for your participation.

Appendix B – REC-H Assessment Form

PG Diploma Cybersecurity – MEOS Application

Section 1 – Project Information

Student Name [PI]	Banele
Student Number [PI]	224612255
Mentor/Supervisor [PRP]	Dr Kativu Kevin

Title of Study

Best Practices for Intrusion Detection and Threat Mitigation in Cloud Environments

Study Type

1.4 This study involves the following (select all that apply):

- ☒ the collection of new data from (or interaction with) one or more individuals through interviews, surveys, focus groups, recordings, etc.
- ☐ access to confidential data or information (or archival data, contact lists or reports), of an organisation (or institution or company) where the data is not available in the public domain (i.e. not available to the general public). The data can be linked to individuals (or clients or employees, etc.).
- ☐ collaboration with an institution (or organisation or company) that is giving access to physical data (or financial data) that is NOT linked to individuals or any personal accounts (or information). Formal access to the data will be requested from an authorised representative of the organisation (or institution or company).
- ☐ access to a database/archive that holds information linked to personal identifiers (e.g. names, ID numbers, account numbers, student numbers); AND/OR the database contains coded information but the researcher has access to the codes that links the information to personal identifiers.
- ☐ the gathering of information/data that is available in the public domain, but that could be regarded as sensitive or potentially sensitive information (e.g., data will be collected via social media networks or public profiles such as Twitter, LinkedIn, Facebook).
- ☐ the collection of data that is accessible in the public domain, is not regarded as sensitive information and is not linked to any individual.
- ☐ the re-use of de-identified data that has already been collected in a previous study (secondary data).

Data Collection

Data Collection

1.5.1 **Recruitment and data collection may only start AFTER written ethics approval has been received.** The Department of Health (2015) prohibits the retrospective review and approval of projects where it is apparent that participant recruitment and data collection commenced prior to ethics approval.

- ☒ I confirm that I have not started with recruitment or data collection.
- ☒ I confirm that I will not start with recruitment or data collection until I have received written ethics approval.

1.5.2 When do you anticipate your data collection to begin? (This date is an indication of your expectations as the Principal Investigator. Data collection may only commence **AFTER** approval has been received. Please select a date that takes into account the ethics review process)

When I obtain
REC-H approval.

1.5.3 The National Health Research Ethics Council stipulates that ethics approval can only be granted for one year. If data collection is envisaged to take more than one year, an extension must be submitted 4-6 weeks prior to the expiry of ethics approval. Only two extensions may be granted, thereafter a new application for ethics approval must be submitted for review.

- ☒ I understand that approval is only given for one year. If data collection is not complete, an application for extension will be submitted prior to ethics approval lapsing.
- ☒ I am aware that the collection of data after ethics approval has lapsed is a breach of NHREC guidelines and will lead to the suspension of the study

Reviewing Faculty/Centre

Reviewing Faculty/Centre

1.6 Select the faculty/centre that you are affiliated with:

Engineering, the Built Environment and Technology

1.6.1 School and Department that the study falls under:

School of Information Technology - Department of IT Management and Governance

SECTION 2 – Risk Assessment

PLEASE NOTE THAT:

- Information is not always collected for research purposes.
- Not all research projects will require formal ethics review and approval.
- Decisions on whether a project requires ethics approval are determined according to its ethics risk classification.
- The REC-H does not give retrospective ethical approval to any project that has already been completed. Thus, it is essential that researchers clarify whether ethical approval for a specific project is required, or not, prior to beginning the data collection.

MANDELA UNIVERSITY RISK ASSESSMENT PROCEDURES:

Procedures for ethics approval are routed according to the level of risk. The Nelson Mandela University REC-H system functions according to the ethical risk classification REGARDLESS of the researcher's level of study. Even undergraduate student projects will require prior review and approval by the REC-H. The level of ethics review is determined by the ethical risks presented by the research project, not the level of degree. The higher the risk, the more in-depth the ethical review process should be. If a less-experienced researcher (e.g., a student at undergraduate or Honours level) chooses a topic that is particularly sensitive, a higher level of scrutiny is required to ensure that the risks to participants and the institution are reduced as far as possible.

Risk is a function of the magnitude of harm and the probability that it will occur.

- Low risk research – where the only foreseeable risk is one of inconvenience or discomfort, or where the probability and magnitude of possible harms implied by participation are no greater than those posed by everyday life in a stable society
- Medium risk research – research involving vulnerable groups, sensitive or potentially sensitive topics, access to private information, and in which there is a potential risk of harm or discomfort, but where appropriate steps can be taken to mitigate or reduce overall risk.
- High risk research poses a real and foreseeable risk of harm or serious adverse consequences to participants and researchers if not managed in a responsible manner.

Please assist us to assess the risk posed by your study by answering the following questions.

2.1 Does your study involve the following? Please select all that apply

- ☐ potential hazards to the physical health (e.g. possibly results in illness, injury, pain) of the participants and/or researcher
- ☐ potential hazards to the psychological or social well-being (e.g. possibly results in feelings of worthlessness, guilt, anger, fear, humiliation, manipulation; social stigmatization; discovery of previously unknown paternity status; disclosure of sensitive and/or sensitive information; anxiety due to being interviewed) of the participants and/or researcher.
- ☐ potential hazards to the legal well-being (e.g. possibly results in the discovery and prosecution of criminal activity) of the participants and/or researcher.
- ☐ potential hazards to the financial well-being of the participants.
- ☐ any process and/or treatment that could be hazardous to the economic well-being and/or result in discomfort associated with the economic well-being of the participants and/or researcher.
- ☐ any process and/or treatment that may result in a traumatic experience for the participants and/or researcher.
- ☐ the potential identification and/or re-identification of a participant from a resulting report or a study in which the anonymity of the participant and/or the confidentiality of the data is not guaranteed AND such identification will present a hazard to participants well-being.
- ☐ any process and/or treatment that could result in risks to non-participants (e.g. distress to relatives upon discovering that a participant suffers from a serious genetic disorder, infectious disease risks to a community, social/economic discrimination of subgroup populations).
- ☒ None of the above

2.2 Does your study involve the following? Please select all that apply

- ☐ vulnerable groups (for example: children under the age of 18 / a sample from an institution (e.g. prison, health facility, place of safety, school, etc.) / persons living with disabilities (e.g. mental, physical, learning) / socially and/or economically disadvantaged persons / persons with limited formal education / the elderly (over 65 years of age) / persons who are not able to or are reluctant to give informed consent (e.g. not being able to understand the language(s) used for the study, but who qualify to participate in the study, undocumented migrants, etc.) / participants undergoing psychological, physiological, or medical testing or treatment, or who are victims of violent crimes / persons who are in a dependency relationship with the Primary Investigator (PI) and/or Primary Responsible Person (PRP) of the study / participants who are a regular part of the PI and/or PRP's social circle / participants who will be financially compensated in any way for participating in the study).
- ☐ Mandela University staff/students
- ☐ the collection and use of human biological samples (e.g. skin, blood, urine, saliva, hair, bones, tumours, and other biopsy specimens) or their exhaled breath.
- ☐ the covert observation (and thus data collection) of behaviour that is not normally in the public domain and in which consent cannot be obtained.
- ☐ the risk of a participant learning about a genetic possibility of developing an untreatable disease.
- ☐ the collection of any articles/documents of property, or of a personal or cultural nature, from participants.
- ☐ the use of specialised equipment on the participants.
- ☐ the use of a questionnaire/survey/interview/focus group/observation practice that collects sensitive data from the participants (e.g. personal data that is not normally in the public domain)?
- ☐ the access of participant data (in individually identifiable or re-identifiable form) from an existing stored repository (e.g. school, institutional, university records or data collected from another previously completed or ongoing research study) or from part of an existing published or unpublished source or database.
- ☐ the potential identification and/or re-identification of a participant from a resulting report or a study in which the anonymity of the participant and/or the confidentiality of the data is not guaranteed.
- ☐ a sponsor who has a vested interest in any possible findings of the study.
- ☐ restrictions or conditions that are attached to the publication and/or presentation of the study results.
- ☒ None of the above

2.3 Does your study involve the following? Please select all that apply

- ☒ human participants (i.e. the collection of new data from (or interaction with) one or more individuals).
- ☐ any process and/or treatment that is expected to result in no foreseeable risk, harm or discomfort to the mental and/or physical well-being of the participants.
- ☒ any process and/or treatment that is expected to result in the only foreseeable discomfort being that of inconvenience (e.g. time and effort required by participants to complete questionnaire, or to participate in a street survey).
- ☐ access to participant information (in non-identifiable form, e.g. summarised form) as part of an existing published or unpublished source or database?
- ☐ access to participant data from an existing stored repository (e.g. school, institutional, university records or data collected from another previously completed or ongoing research study) that relies exclusively on publicly available information.
- ☐ access to participant data from an existing stored repository (e.g. school, institutional, university records or data collected from another previously completed or ongoing research study) that relies exclusively on secondary use of anonymous / anonymised / de-identified information (i.e. no identifiable information is generated or inferred).
- ☐ research that requires an ethics reference number for publication purposes or for examination
- ☐ None of the above

2.4 Does your study involve the following? Please select all that apply

- ☐ research that DOES NOT involve humans or use data or biological materials belonging to humans.
- ☐ research that relies exclusively on information that is publicly available (i.e., involves the analysis of information in the public domain) or that is accessible in terms of legislation or regulation, and (1) does NOT require gatekeeper, site, or platform permission, (2) does NOT involve sensitive or potentially sensitive information, (3) AND uncontroversial topics are being investigated, (4) AND participants are not from vulnerable populations, (5) AND data collection and dissemination of research findings does not identify individuals or groups and does not create the possibility of re-identification.
- ☐ research that involves the observation of people in public spaces (including virtual public spaces) and natural environments, provided that: (1) the researcher does not interact directly with individuals or groups, (2) the researcher does not stage any intervention, (3) the individuals or groups do not have a reasonable expectation of privacy, and (4) dissemination of research findings does not identify individuals or groups and does not create the possibility of re-identification.
- ☐ research that is exclusively for quality assurance, quality improvement (audits), programme evaluation activities, performance reviews, and consumer surveys.
- ☒ None of the above

Appendix C – Cover Page of Turnitin Report

Mlonyeni 224612255 Final Research Report 2025.pdf

ORIGINALITY REPORT

19%

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